

Autodesk® **Scaleform**®

Scaleform 4.3

Scaleform 4.3

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Autodesk® Scaleform® 4.3

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How to Contact Autodesk Scaleform:

Document	Scaleform 4.3 Renderer Threading Guide
Address	Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

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Scaleform 4.0

(Advance)

Scaleform SDK

Scaleform 4.0 API

Scaleform

2

API Scaleform3.x Scaleform4

1. Renderer2D
2. GfxLoader
- 3.
4. MovieDisplayHandle
Renderer2D::Display
5. Gfx::Movie
6. API
Render::HAL
GRenderer Scaleform 3.X

2.1

Scaleform Render::Renderer2D Render::HAL Renderer2D
Display Render::HAL Renderer2D
" " Render::D3D9::HAL

Render::PS3::HAL

```
D3DPRESENT_PARAMETERS   PresentParams;  
IDirect3DDevice9*        pDevice = ...;  
ThreadId                renderThreadId = Scaleform::GetCurrentThreadId();  
Render::ThreadCommandQueue *commandQueue = ...;  
Ptr<Render::D3D9::HAL>   pHal;  
Ptr<Render::Renderer2D> pRenderer;  
  
pHal = *SF_NEW Render::D3D9::HAL(commandQueue);  
if (!pHal-> InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
```

```

renderThreadId)))
{
    return false;
}
pRenderer = *SF_NEW Render::Renderer2D(pHal);

```

HALInitParams class

pDevice PresentParams

renderThreadId

ThreadCommandQueue

Renderer2D HAL

2.2

Scaleform4

Renderer2D

Scaleform 3.x

Loader

InitHAL

Render::HAL

ShutdownHAL

Render::HAL

HAL

```

MeshCacheParams mcp;
mcp.MemReserve      = 1024 * 1024 * 3;
mcp.MemLimit        = 1024 * 1024 * 12;
mcp.MemGranularity  = 1024 * 1024 * 3;
mcp.LRUTailSize     = 1024 * 1024 * 4;
mcp.StagingBufferSize = 1024 * 1024 * 2;

pRenderer->GetMeshCacheConfig()->SetParams(mcp);

```

MemReserve

MemLimit

MeshCache

MemGranularity

LRU

LRUTailSize

MeshCache

StagingBufferSize

64K

PC

```

GlyphCacheParams gcp;
gcp.TextureWidth = 1024;
gcp.TextureHeight = 1024;
gcp.NumTextures = 1;
gcp.MaxSlotHeight = 48;

pRenderer->GetGlyphCacheConfig()->SetParams(gcp);

```

Scaleform 3.3 TextureWidth TextureHeight NumTextures
MaxSlotHeight

InitHAL InitHAL
SetParams

2.3

Scaleform 4.0 Gfx::ImageCreator Render::TextureManager
Scaleform ImageCreator
Gfx::Loader

```

Gfx::Loader loader;
...

SF::Ptr<Gfx::ImageCreator> pimageCreator =
    *new Gfx::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);

```

ImageCreator TextureManager TextureManager
Render::HAL::GetTextureManager()
GetTextureManager

Render::HAL TextureManager CreateTexture

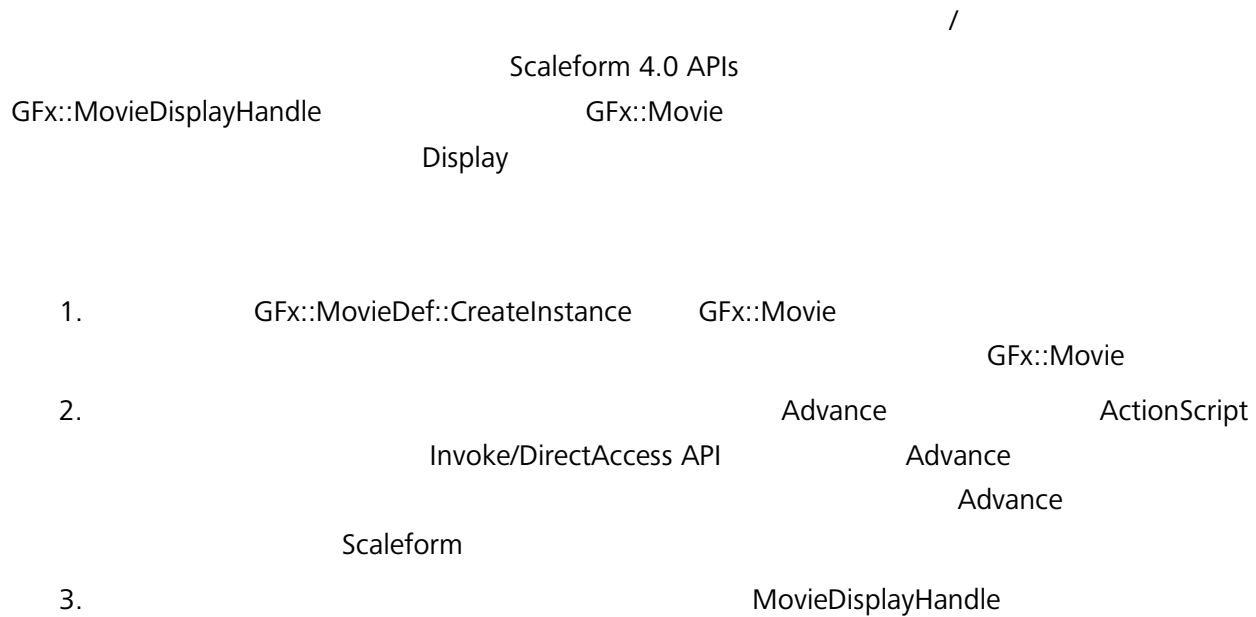
D3D9 D3DCREATE_MULTITHREADED

- D3DCREATE_MULTITHREADED D3D9 ThreadCommandQueue Render::HAL

D3D ThreadCommandQueue

Platform::RenderHALThread

2.4



```
//-----
// 1.
Ptr<GfX::Movie> pMovie = ...;
GfX::MovieDisplayHandle hMovieDisplay;

//
//
pMovie->SetViewport(width, height, 0,0, width, height);
hMovieDisplay = pMovie->GetDisplayHandle();

// Scaleform Player
//
pRenderThread->AddDisplayHandle(hMovieDisplay);
```

```

//-----
// 2.

//
//   ExternalInterface      Advance
//   float deltaT = ...;
//   pMovie->HandleEvent(...);
//   pMovie->Advance(deltaT);

//
//
//   pRenderThread->WaitForOutstandingDrawFrame();
//   pRenderThread->DrawFrame();

//-----
// 3.      -      DrawFrame
//   Ptr<Render::Renderer2D> pRenderer = ...;

//   pRenderer->BeginFrame();
//   bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());
//   if (hasData)
//       pRenderer->Display(hMovieDisplay);
//   pRenderer->EndFrame();

```

- WaitForOutstandingFrame – CPU
- Advance
- MovieDisplay.NextCapture: “ “Advance false

2.5

Gfx::Movie Release

Render::ThreadCommandQueue Movie
ThreadCommandQueue

Movie::ShutdownRendering(false)
Movie::IsShutdownRenderingComplete
NextCapture false

3

Scaleform

—

GFX::Movie::Advance

ActionScript

Advance

Advance

Scaleform

Advance

Render::HAL

Scaleform

•

•

Scaleform

Player

Scaleform

3.1

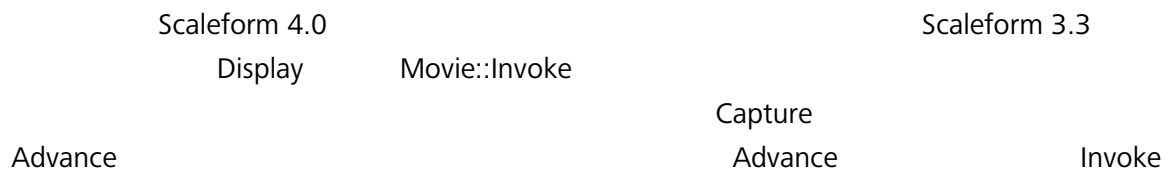
Advance

/

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GFX::Movie::Advance

- Movie::Advance
- Capture NextCapture Capture
- Capture Advance Invoke Direct Access API Movie



3.2

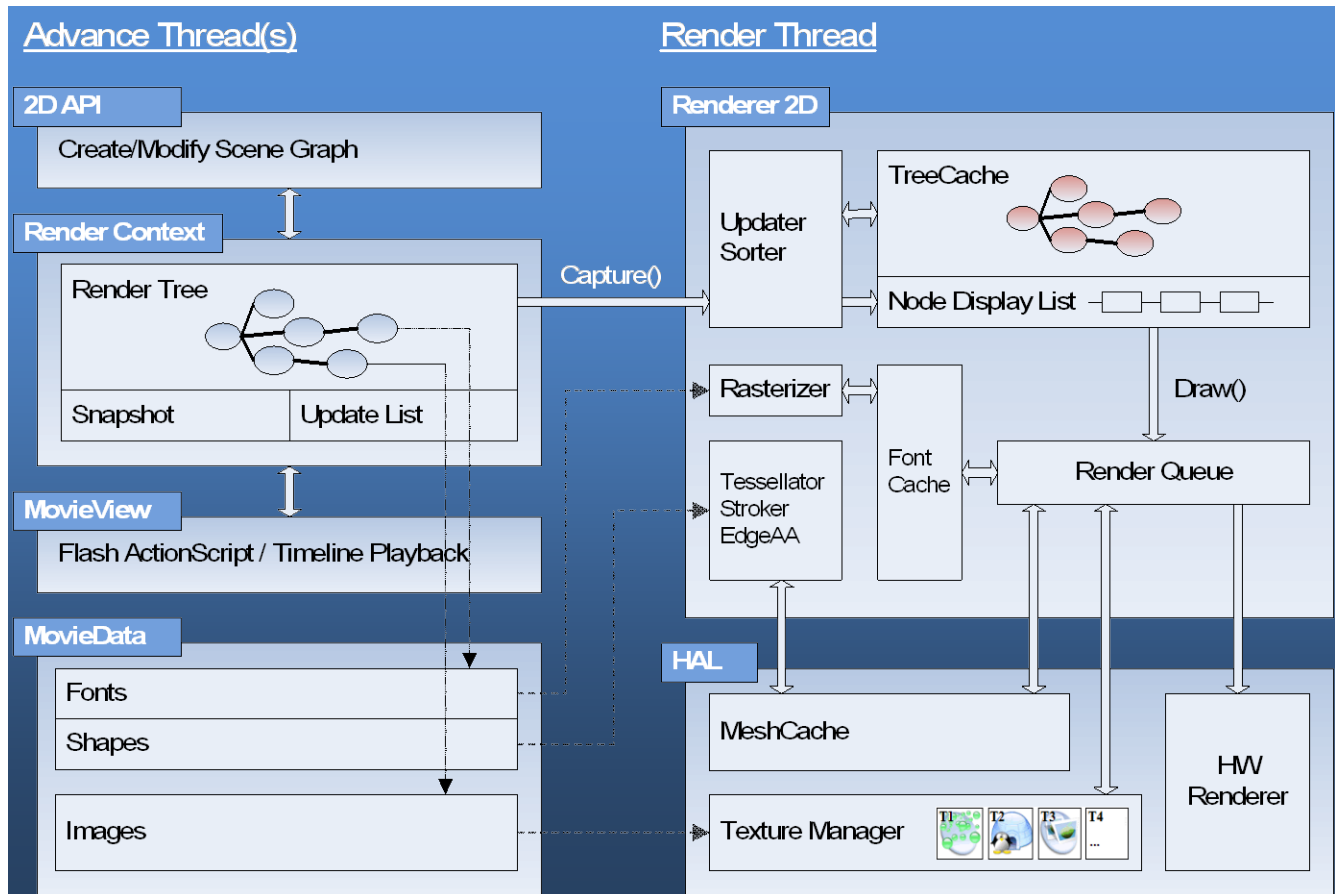
Render::Render2D::Display

Render::Render2D::Display

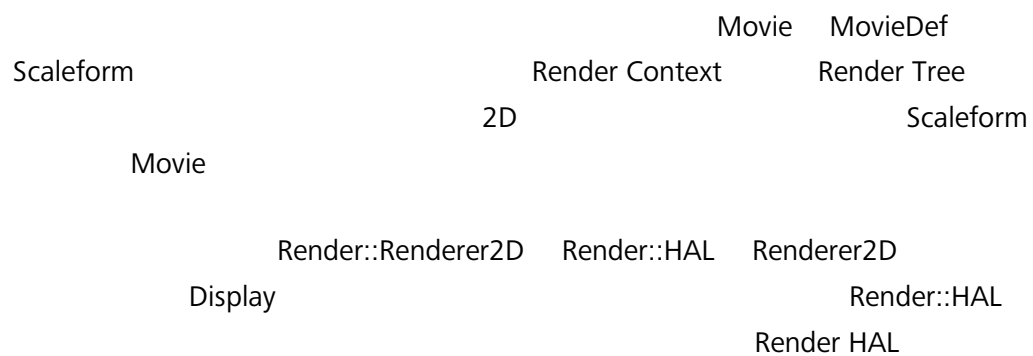
Scaleform

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1

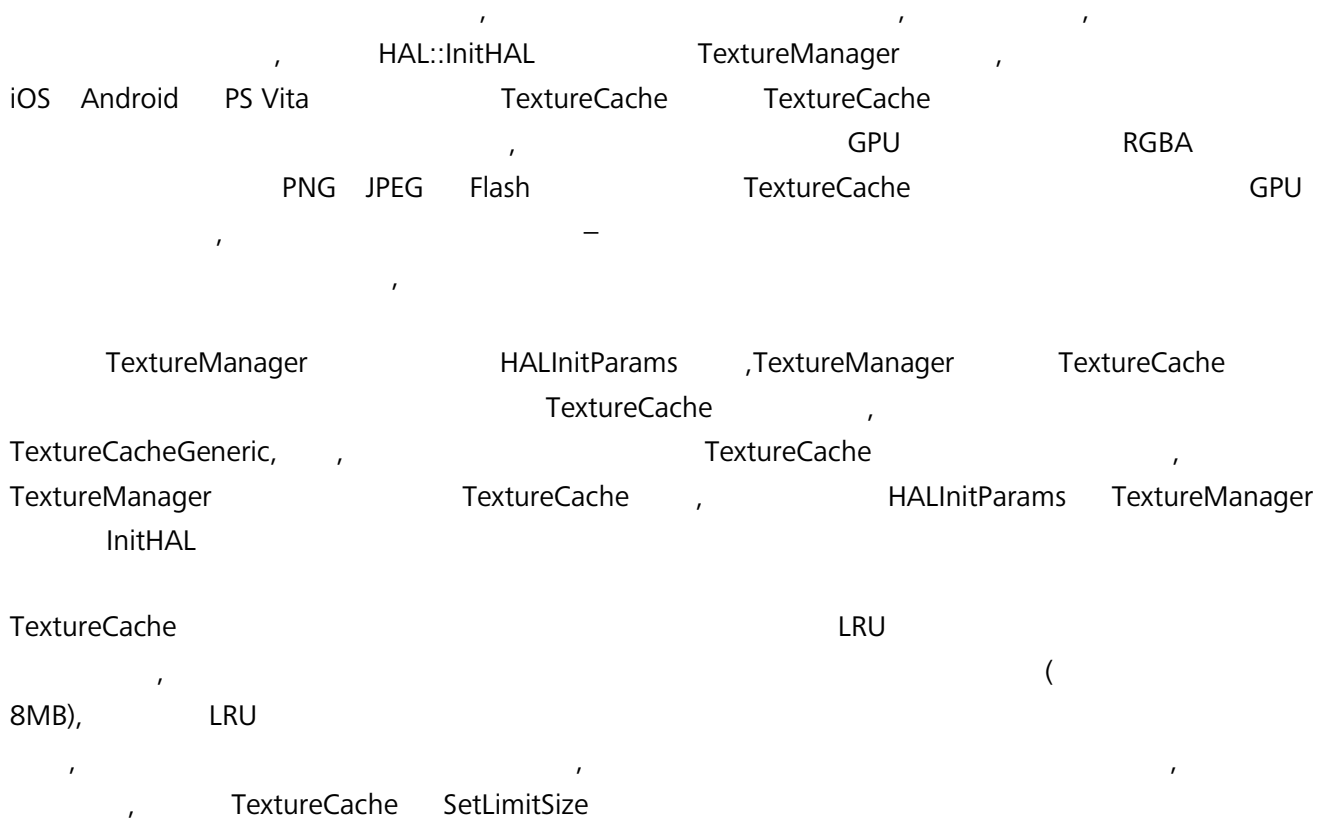


-
- The diagram illustrates the rendering pipeline components and their interactions:
- Font Cache**: Located at the top left and top right.
 - Mesh Cache**: Located below the Font Cache on the left.
 - Render Queue**: Located in the center, with two instances labeled "Render Queue".
 - HAL API**: Located at the bottom left.
 - ActionScript**: Located at the bottom center.
 - AI**: Located at the bottom right.
 - CPU**: Located on the far right.
- Arrows indicate the flow of data and control:
- From **Font Cache** to **Render Queue**.
 - From **Mesh Cache** to **Render Queue**.
 - From **Render Queue** to **HAL API**.
 - From **HAL API** to **ActionScript**.
 - From **ActionScript** to **AI**.
 - From **AI** to **CPU**.
 - From **CPU** to **Font Cache**.

5 渲染子系统说明

5.1 TextureCache

TextureCache



Position2dOffset

Position2dOffset

```
<ShaderSource id="Position2dOffset" pipeline="Vertex">
```

```
    Position2d(
    attribute float4 pos      : POSITION,
    varying   float4 vpos    : POSITION,
    uniform   float4 mvp[2],
    uniform   float  poffset)
    {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
    }
</ShaderSource>
```

Position2d - 'poffset' x y

, D3D9_Shader.cpp , ShaderInterface::SetStaticShader

```
//
    if (pHal->UserDataStack.GetSize() > 0 )
    {
        const UserDataState::Data* data = pHal->UserDataStack.Back();
        if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
            data->RendererString.CompareNoCase("abc") == 0)
        {
//
            ,
            vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
            shader = (FragShaderDesc::ShaderType) ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
        }
    }
}
```

HAL UserDataStack (abc

,) , Position2dOffset

(D3D9_ShaderDescs.h)

ShaderData.xml ,

'poffset'

ShaderInterface::Finish

//

```

if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
    {
//    poffset
        for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
        {
            float poffset = data->RendererFloat / 256.0f;
            SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
        }
    }
}

,          'meshCount'
          (
1          HAL          ,          'm'
          ),meshCount

```

6.3

```

HAL          ,          Scaleform
          ,          ,          Scaleform
          ,          Scaleform
          /          ,          AS2
disableBatching          AS3  disableBatching          AS2/AS3
          ,

```


,HAL

GfxShaderMaker '-featurelevel'

GfxShaderMaker.exe -platform D3D1x -featurelevel
FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1

	FEATURE_LEVEL9_3		FEATURE_LEVEL_9_3
InitHAL,		(	FEATURE_LEVEL_9_1)